

# CRISTINA CUESTA BRET

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Early-career Mechanical Engineer with hands-on experience in mechanical design and SolidWorks. Passionate about renewable/sustainable technologies, innovating in mechanical design, and applying cutting-edge solutions to enhance project efficiency and user experience.

## EDUCATION

### Bachelor of Science in Mechanical Engineering

#### Certificate in Artificial Intelligence Fundamentals and Applications

University of Florida, Gainesville

Overall & Major GPA: 3.53 & 3.61

Aug. 2021 – May 2026

**Relevant Coursework:** Thermodynamics, Fluid Mechanics, Energy Conversion, Controls, Materials, FEA, Manufacturing, Engineering Leadership

## SKILLS

**Software Skills:** SolidWorks, MATLAB, Python, Microsoft Office, Granta, PDM, Abaqus, MS Excel

**Technical Skills:** 3D CAD software, manufacturing engineering, FEA, programming, technical writing, engineering drawings, materials

**Languages:** Spanish (Native), English (Fluent)

## WORK EXPERIENCE

### Teaching Assistant

University of Florida

Aug. 2023 – Present

Gainesville, FL

- Assisted in multiple courses, providing lab instruction, and assignment grading and feedback, serving as a key resource for technical guidance and professional development.
- Mentored and supported 100+ undergraduate students through experiments, circuit building, and data analysis while improving their technical report writing and engineering communication.
- Dynamics and Controls Systems Design Laboratory (Fall 2025).** Supervised 6 weekly labs using a Furuta Pendulum; provided feedback on transfer functions, bang-bang, PID, and full-state feedback control as well as LabVIEW coding.
- Mechanics of Materials Laboratory (Spring 2025).** Led 4 weekly labs, mentoring students on tensile testing, strain gauges, pressure experiments, and LabVIEW coding; assessed 60+ lab reports with detailed feedback.
- Elements of Electrical Engineering (Fall 2023 – Fall 2024).** Reinforced fundamental electrical engineering concepts and circuit building.

### Mechanical Engineering Summer Intern

GMV Aerospace and Defence SAU

Jul. 2025 – Aug. 2025

Madrid, Spain

- Collaborated with senior engineers on the CAD and assembly of electronic hardware systems for defense and aerospace applications.
- Independently executed a full design project using SolidWorks, collaborating with the production team to adapt to evolving constraints and deliver a manufacturable solution.
- Operated and maintained 3D resin printers to validate and refine prototypes through iterative assessments.
- Supported structural, vibration, and shock analyses on 2 mechanical components to ensure reliability under operational loads.
- Executed material selection analyses for 2 projects, balancing mechanical performance and application constraints.

### Undergraduate Research Intern

Universitat de Lleida

May 2023 – July 2023

Lleida, Spain

- Developed a sophisticated 700-line solar management Python algorithm aimed at optimizing the allocation of Direct Normal Irradiance (DNI) resources and providing accurate power output forecast based on dynamic demands and boundary conditions.
- Streamlined Python code to support various receiver sizes and electrical configurations of photovoltaic (PV) cells, improving the tool's versatility and performance by 20%.

## INVOLVEMENT

### Chassis and Cockpit Engineer

Solar Gators

Jan. 2023 – Present

Gainesville, FL

- Executed multiple hands-on manufacturing tasks, including carbon fiber and resin wet layups, to support vehicle production.
- Designed and manufactured an actuator release body latch system and canopy hinge using SolidWorks contributing to the structural and functional integrity of the car while following FSGP and ASC regulations.
- Designed the steering wheel for the new solar car, implemented LCD and paddle shifters for regenerative braking.

### President, Secretary and Social Media Manager

University of Florida Club Tennis

May. 2022 – Present

Gainesville, FL

- First female** to lead the University of Florida Club Tennis team with +150 members.
- Manage home and away tournaments with +30 teams.
- Directed the design of team uniforms and increased sales by 60%. Managed social media accounts with +2,500 followers, increasing engagement and impressions by 80%.

**Honors:** Herbert Werheim College of Engineering **Dean's List**, **Deputy Lead** in Senior Mechanical Design Capstone, **Honors Thesis**

**Certifications:** SolidWorks, CNC Machining Training Program, Cambridge Proficiency in English (CPE)